5. + hadron decays (what the detector will see)

standalone hadron decays: event_Decay.lhe

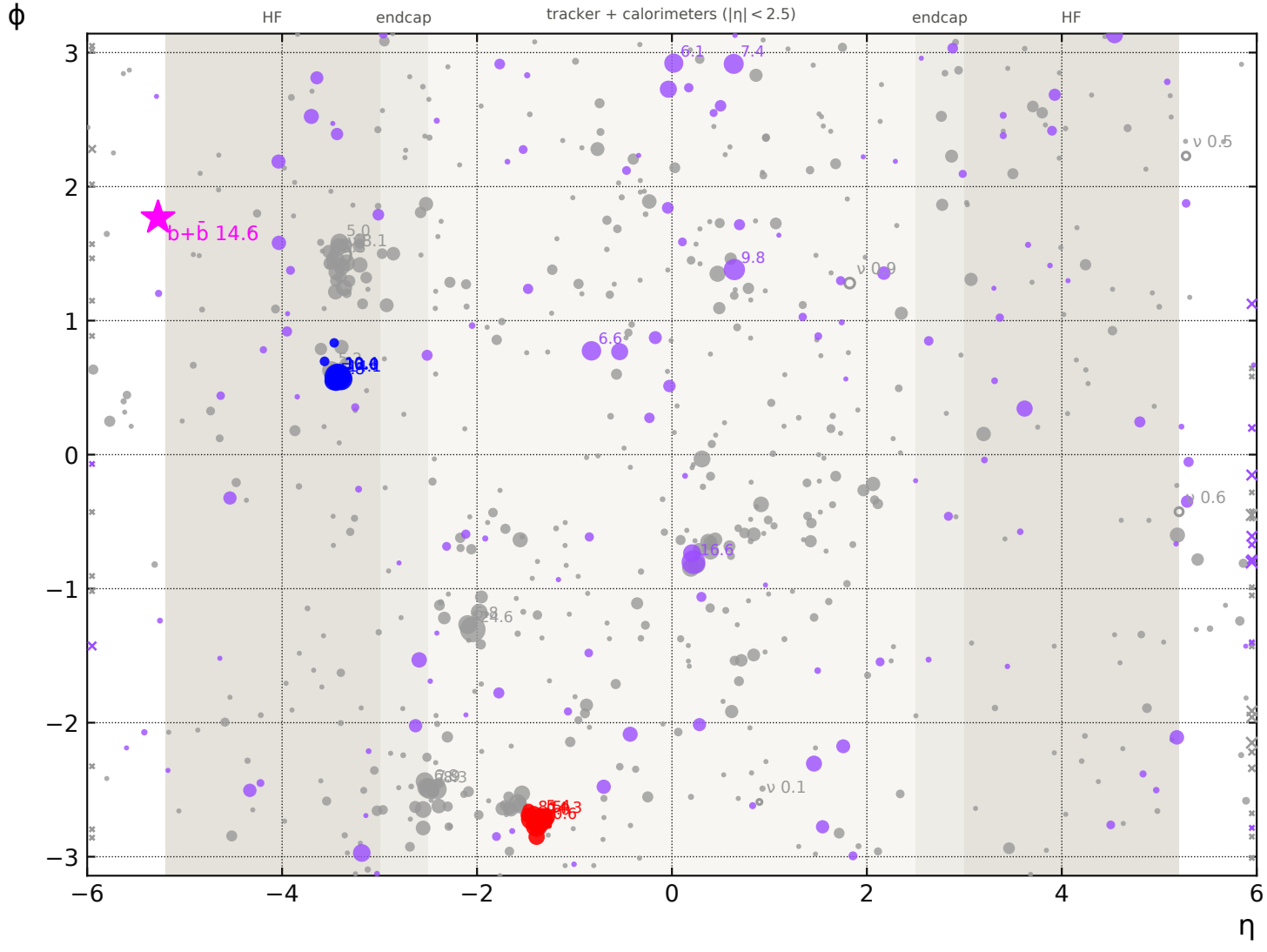
749 particles drawn (marker size ~ log p_T; label = p_T [GeV] for $p_T > 5$ GeV), 43 with $|\eta| > 6$ drawn on the border (x)

b hadron $\bar{B}^{*0}$ decay products (n=16): p_T = 78.56 GeV, η = -1.38, ϕ = -2.73; the $\bar{B}^0$ flew L = 21.0 mm (L_{xy} = 9.9 mm)

$\bar{b}$ hadron B^{*0} decay products (n=10): p_T = 82.83 GeV, η = -3.43, ϕ = 0.58; the $\bar{B}^0$ flew L = 146.9 mm (L_{xy} = 9.5 mm)

net b+$\bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})|$ = 14.57 GeV (p_x = -2.85, p_y = 14.28), $\Sigma|p_T|$ = 161.39 GeV, $\Delta\phi(b, \bar{b})$ = 2.97

all 749 final particles: $|\Sigma\vec{p}_T|$ = 0.02 GeV (decays create no momentum); 5 neutrinos carry 229.4 GeV, missing p_T = 9.13 GeV



particle origin (number drawn)

- b system (16)
- $\bar{b}$ system (10)
- b + $\bar{b}$ (vector sum)
- primary hadron (143)
- decay product (580)
- neutrino (hollow, ν)

33 colour-singlet strings

marker size:

- p_T = 1 GeV
- p_T = 10 GeV
- p_T = 100 GeV

stages:

1 hard process
2 ISR + FSR
3 MPI / remnants
4 hadronisation
5 hadron decays